ALGOPILOT
(Algorithm Visualizer and Execution Platform)

Group 05 | Shikshita Yadav • Disha Waghmare • Pratiksha Pathak

Guide: Dr. Jaya Gupta

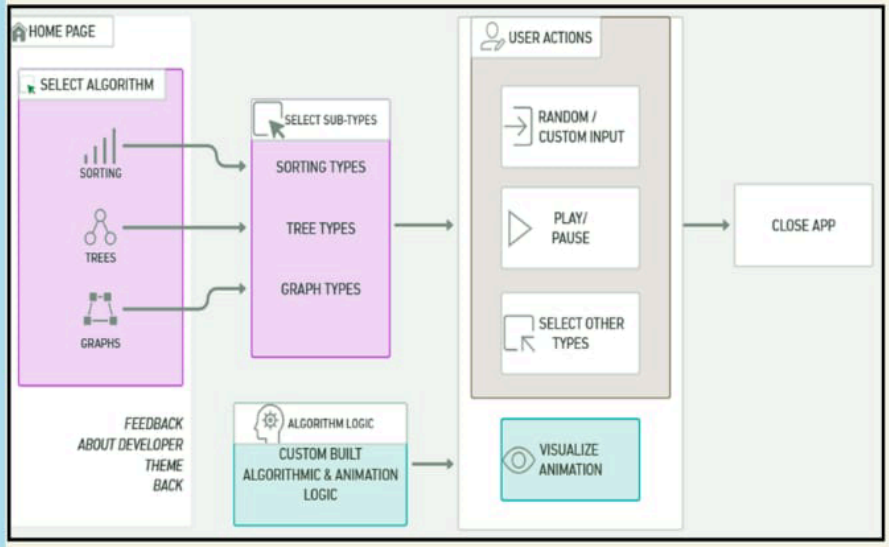
Abstract

- AlgoPilot is an Android app that visualizes DSA algorithms step-by-step using animations and pseudocode.
- It enables interactive, offline, and self-paced learning to improve algorithm understanding.

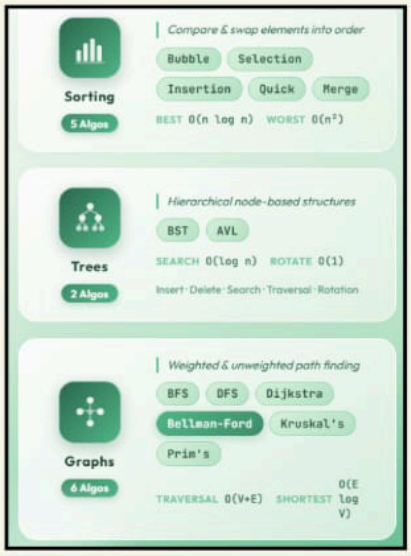
Technologies Used

Android Studio | Object Animator | Gradle | XML | Java

Application Architecture



Algorithms Implemented



Application Results

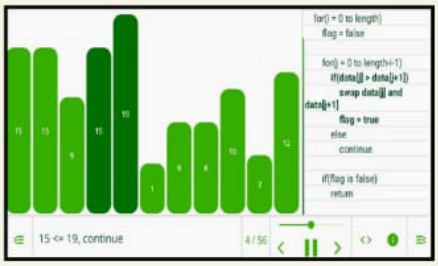


Fig. 1. Sorting Example

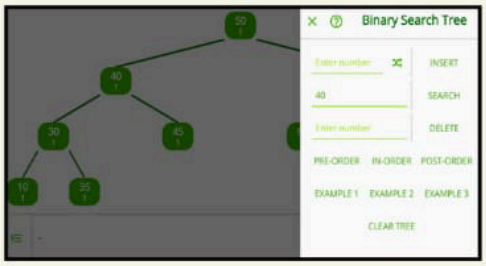


Fig. 2. Tree Example

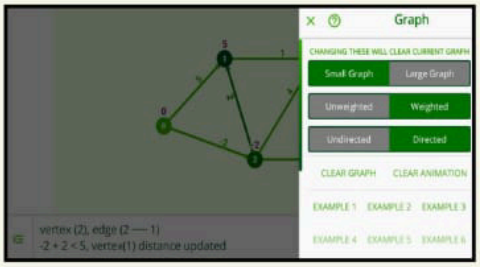
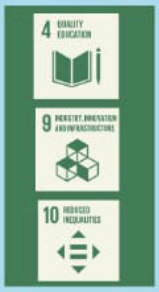


Fig. 3. Graph Example

SDG Goals



Conclusion

- Visualizes algorithms using animations and synchronized pseudocode, helping students clearly understand the step-by-step logic behind Data Structures and Algorithms.
- Makes DSA learning interactive and mobile-friendly, allowing students to study anytime and anywhere through an Android application.
- Supports self-paced learning with controls to adjust execution speed while observing algorithm behavior.

References

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- Sharma, P., Verma, K.: The Efficacy of Animation and Visualization in Teaching Data Structures: A Case Study (2024)
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